

**1. O/N/23/21/Q5**

Most of the energy emitted by the Sun is in three regions of the electromagnetic spectrum.

**(a)** The name of one of these three regions is the ultraviolet region.

**(i)** State the names of all three regions in order of increasing frequency.

smallest frequency .....

intermediate frequency .....

largest frequency .....

[2]

**(ii)** Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect .....

.....

property .....

.....

[2]

**(b)** Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

2. M/J/23/22/Q7

(a) Ultraviolet radiation is one component of the electromagnetic spectrum.

(i) State the name of **two** components of the electromagnetic spectrum that have a smaller wavelength than ultraviolet radiation.

1 .....

2 ..... [1]

(ii) State **one** useful application of ultraviolet radiation.

..... [1]

(iii) Exposure to ultraviolet radiation from the Sun damages the eyes.

State **one** type of damage to the eye caused by ultraviolet radiation.

..... [1]

(b) Fig. 7.1 shows a ray of light. The ray passes into a semi-circular block of glass at A and leaves the glass at B, travelling along the surface to C.

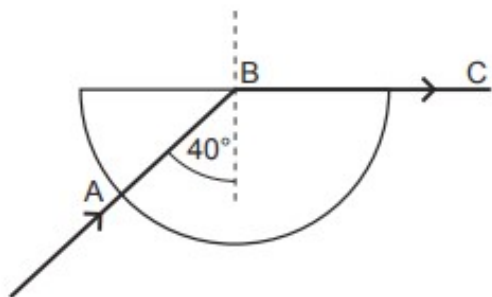


Fig. 7.1

(i) State the name given to the angle of incidence marked as 40°.

..... [1]

(ii) Calculate the refractive index of the glass.

refractive index = ..... [2]

[Total: 6]

3. O/N/22/21/Q4

X-rays are transverse waves that are part of a spectrum of waves.

- (a) State the name of the spectrum of waves that includes X-rays.

..... [1]

- (b) (i) Explain what is meant by 'frequency'.

.....

.....

..... [2]

- (ii) State the name of the waves in this spectrum that have the greatest frequency.

..... [1]

- (c) X-rays are used in hospitals to produce an image of broken bones.

- (i) Explain how this is done. You may draw a diagram if you wish.

.....

.....

.....

..... [3]

- (ii) Explain why X-rays are **not** used for pre-natal scanning.

.....

..... [1]

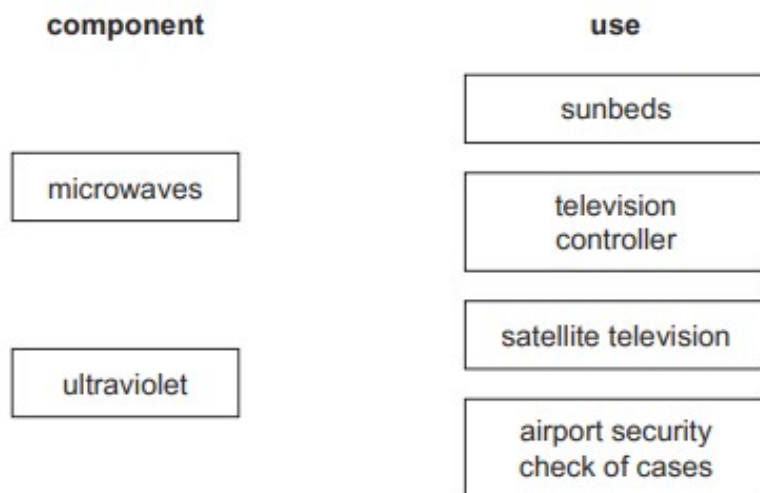
[Total: 8]

#### 4. M/J/20/22/Q5

The components of the electromagnetic spectrum have different uses.

Microwaves are used in cooking and ultraviolet rays are used in sterilisation.

- (a) Draw **one** line from each component of the spectrum to another suitable use for that component.



[2]

- (b) Fig. 5.1 shows a microwave oven used to heat soup. The container for the soup is a glass bowl.

Microwaves created inside the oven are reflected by the metal walls.



Fig. 5.1

- (i) Explain why the choice of material for the container is important in microwave cooking.

.....

.....

..... [1]

5. O/N/19/21/Q4

Fig. 4.1 represents the electromagnetic spectrum.

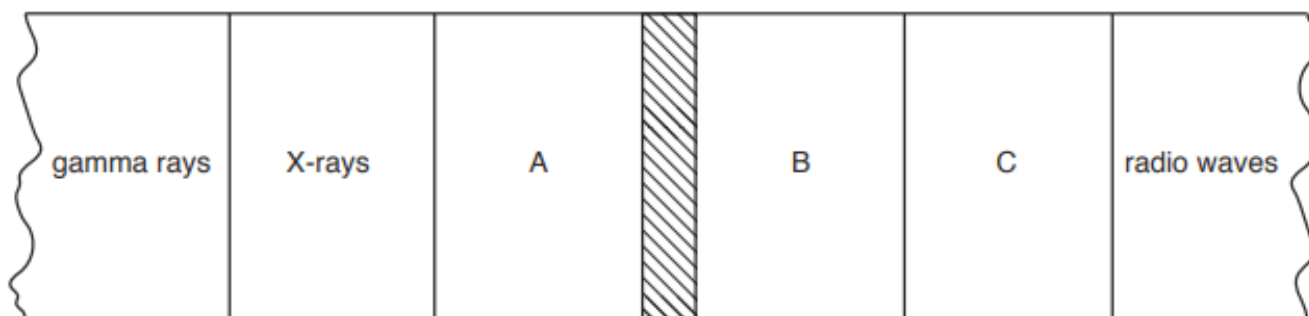


Fig. 4.1

The shaded region in the middle is the visible light spectrum.

(a) State the name of the component labelled:

- A .....
- B .....
- C .....

[2]

(b) State the name of the component of the electromagnetic spectrum that has:

the greatest frequency .....

the smallest wavelength .....

[1]

(c) Explain how X-rays are used to produce an image of a broken bone in a patient's leg.

.....  
.....  
.....  
.....  
.....

[3]

[Total: 6]

6. M/J/19/22/Q4

A star emits electromagnetic radiation over a range of wavelengths.

Fig. 4.1 shows the brightness of the radiation from the star at different wavelengths.

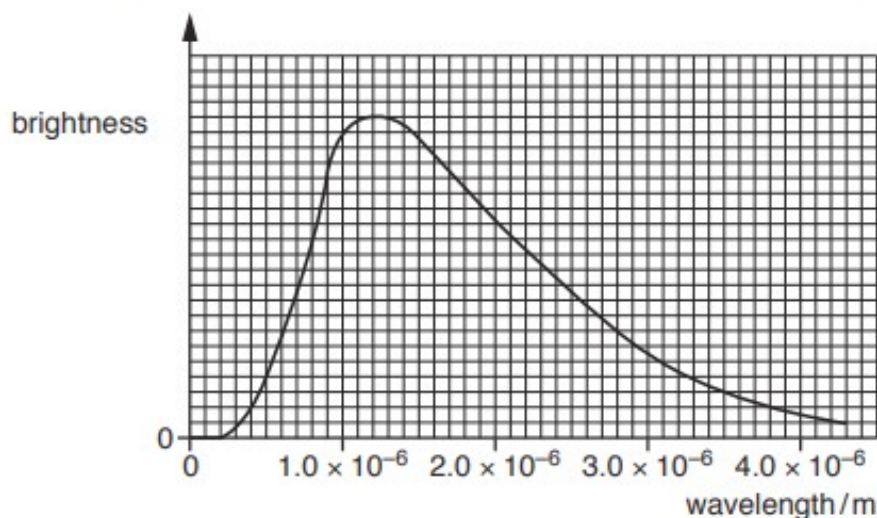


Fig. 4.1

Radiation from the star is brightest at one wavelength.

- (a) Determine the wavelength where the radiation is brightest.

wavelength = ..... [1]

- (b) Visible light has a wavelength between  $4.0 \times 10^{-7}$  m and  $7.0 \times 10^{-7}$  m.

The radiation in (a) lies just outside the visible part of the electromagnetic spectrum.

State the name of the region of the spectrum that contains the radiation in (a).

..... [1]

- (c) Determine the frequency of the radiation in (a).

The speed of light is  $3.0 \times 10^8$  m/s.

frequency = ..... [3]

[Total: 5]



7. O/N/17/21/Q10

Fig. 10.1 represents the electromagnetic spectrum.

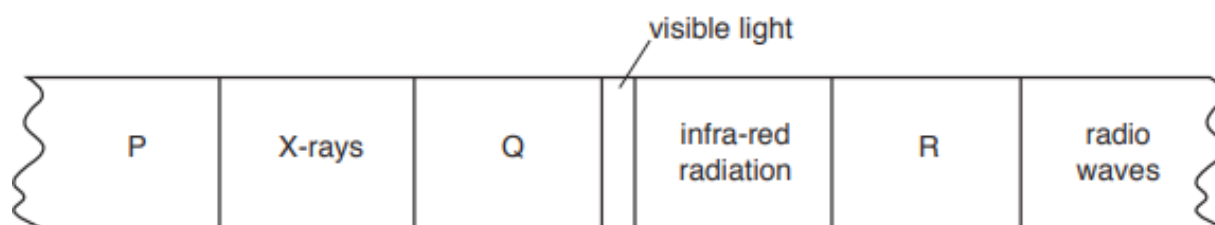


Fig. 10.1

(a) State the name of the component of the electromagnetic spectrum that is found in region

(i) P, .....

(ii) Q, .....

(iii) R, .....

[2]

(b) On Fig. 10.1, mark a tick (✓) in all of the boxes that represent the components with frequencies greater than the frequencies of visible light. [1]

(c) The speed of electromagnetic radiation in a vacuum is  $3.0 \times 10^8 \text{ m/s}$ .

A television remote controller uses infra-red radiation of wavelength  $9.4 \times 10^{-7} \text{ m}$ .

(i) Calculate the frequency of this radiation.

frequency = ..... [3]

(ii) Explain how infra-red radiation is used in television remote controllers.

.....  
 .....  
 .....  
 ..... [2]

8. M/J/16/22/Q6

Fig. 6.1 shows some of the colours in the visible part of the electromagnetic spectrum.

|     |        |        |       |      |
|-----|--------|--------|-------|------|
| red | orange | yellow | green | blue |
|-----|--------|--------|-------|------|

**Fig. 6.1**

(a) State which colour in Fig. 6.1 has

(i) the largest wavelength, ..... [1]

(ii) the highest frequency. .... [1]

(b) Visible light, infra-red, ultra-violet and radio waves are four components of the electromagnetic spectrum.

State two other components of the electromagnetic spectrum and describe a different use that is made of each component.

1. component .....

description of use .....

.....

.....

2. component .....

description of use .....

.....

.....

[4]



9. M/J/13/22/Q4

- (a) The list below contains three components of the electromagnetic spectrum.

infra-red

gamma rays

visible light

Arrange the components in order of increasing wavelength.

..... [1]

- (b) Satellites are used in the transmission of some television signals.

Fig. 4.1 shows a satellite above the television station where a television signal is generated.

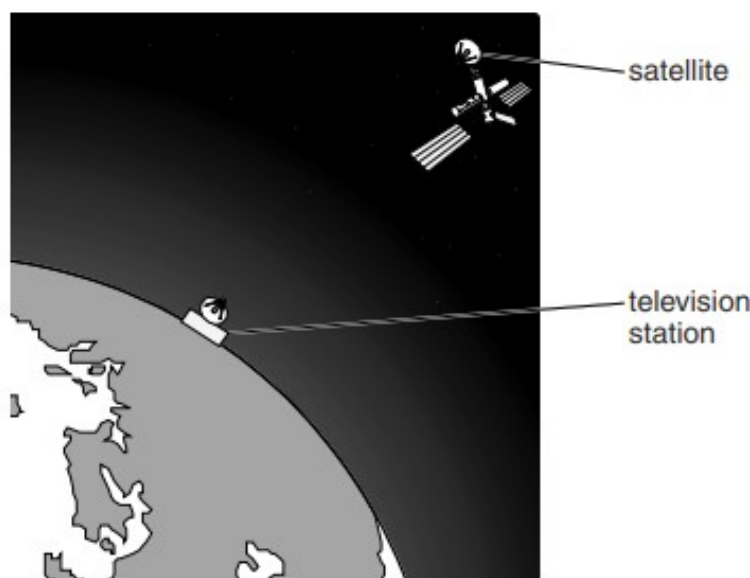


Fig. 4.1 (not to scale)

- (i) State which component of the electromagnetic spectrum is used to transmit the television signal to the satellite.

..... [1]

- (ii) Explain how the satellite is used.

.....

..... [1]

- (iii) Suggest one advantage of using a satellite to transmit television signals.

.....

..... [1]

10. O/N/12/21/Q6

(a) Fig. 6.1 shows part of a page from a pupil's notebook.

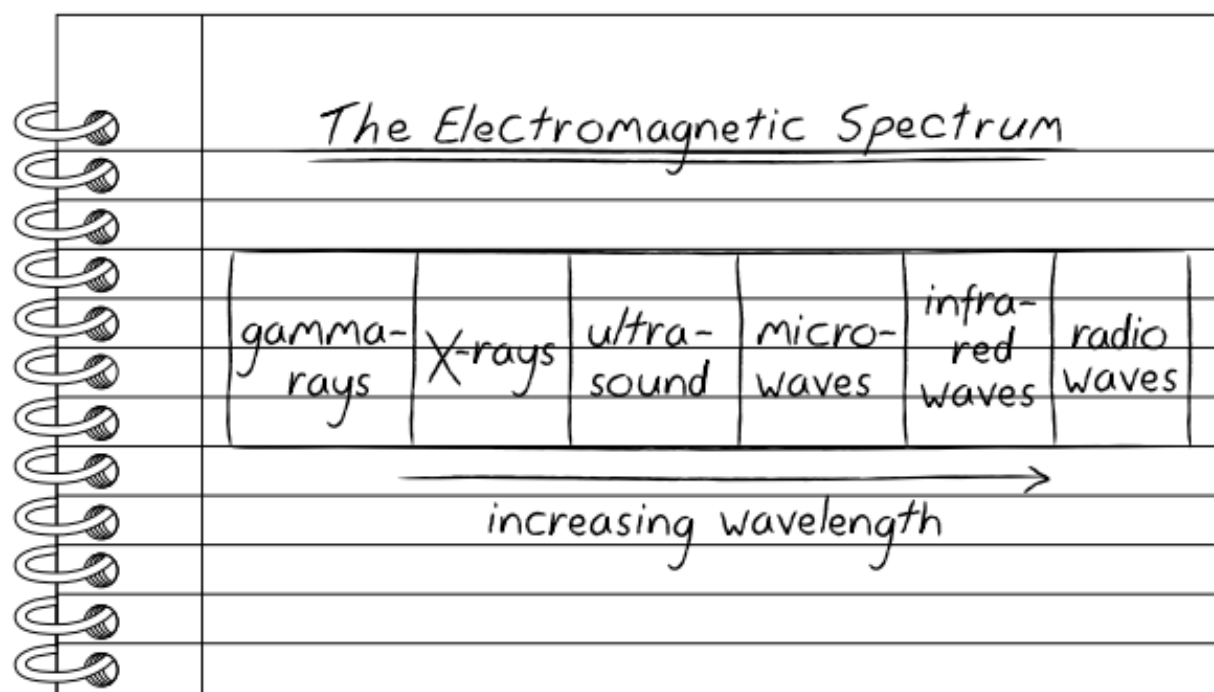


Fig. 6.1

List three errors in the pupil's notes.

1. ....
2. ....
3. ....

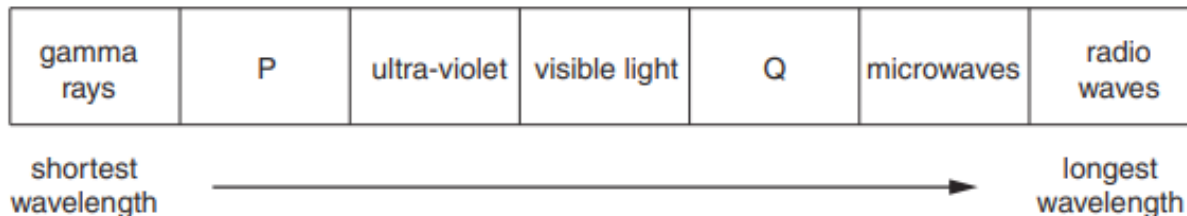
[3]

(b) State and explain one application of X-rays in engineering.

- .....
- .....
- ..... [2]

11. M/J/12/21/Q5

Fig. 5.1 shows the electromagnetic spectrum.



**Fig. 5.1**

Two components of the spectrum, P and Q, have not been named.

**(a)** State the name of

**(i)** component P,

.....[1]

**(ii)** component Q.

.....[1]

**(b)** All electromagnetic waves have a wavelength and a frequency. They all have changing magnetic and electric fields. State two other properties of all electromagnetic waves.

1. ....

.....

2. ....

.....

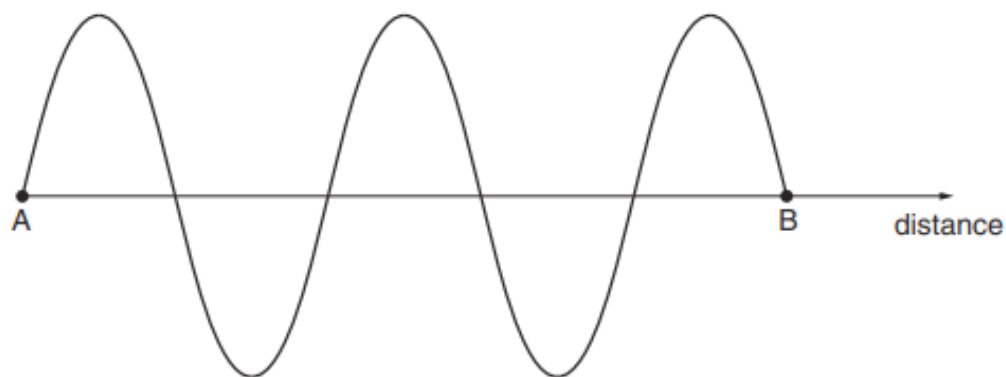
[2]

**(c)** State the component of the electromagnetic spectrum used for satellite communication.

.....[1]

12. O/N/11/22/Q4

Fig. 4.1 represents a microwave travelling in air through points A and B.



**Fig. 4.1** (not to scale)

The distance between A and B is 40 cm.

- (a) State the speed of microwaves in air.

speed = .....[1]

- (b) Determine the wavelength of the microwave shown in Fig. 4.1.

wavelength = .....[1]

- (c) Describe how microwaves are used in the transmission of television signals by satellite.

.....  
 .....  
 .....  
 .....[3]

- (d) State two properties common to all electromagnetic waves.

1. ....  
 2. .....  
 .....[2]

**13. M/J/08/Q5**

- (a) Each object in the table below emits one main type of electromagnetic wave. Complete the table by writing in the name of the type of wave. One line has been written for you.

| Object                          | Main type of electromagnetic wave emitted |
|---------------------------------|-------------------------------------------|
| radio transmitter               | radio wave                                |
| remote control for a television |                                           |
| radioactive source              |                                           |

[2]

- (b) X-rays are used in hospitals to produce images of bones and to show whether bones are broken.

- (i) State what is used to detect X-rays.

.....  
 ..... [1]

- (ii) Explain the properties of X-rays that enable an image of a bone to be produced.

.....  
 .....  
 .....  
 .....  
 .....  
 ..... [2]

**14. O/N/05/Q4**

X-rays, microwaves, ultra-violet rays and infra-red rays are different types of radiation in the electromagnetic spectrum.

- (a) Write the name of one of these types of radiation in each of the boxes, placing them in order of increasing wavelength.

shortest wavelength  longest wavelength

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

[2]

- (b) State one use of ultra-violet radiation.

.....

.....

..... [1]

- (c) State two properties that are common to all types of radiation in the electromagnetic spectrum.

1. ....

.....

2. ....

..... [2]